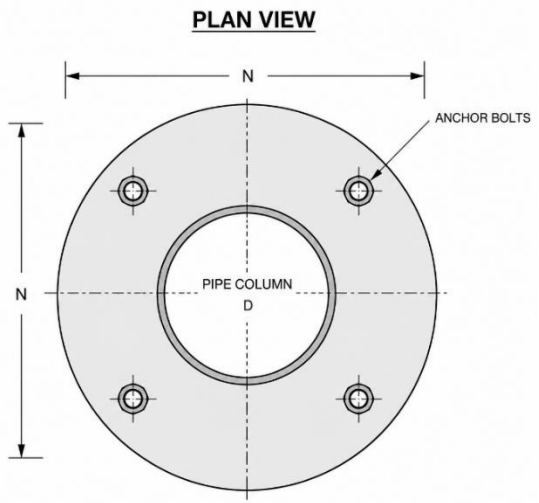
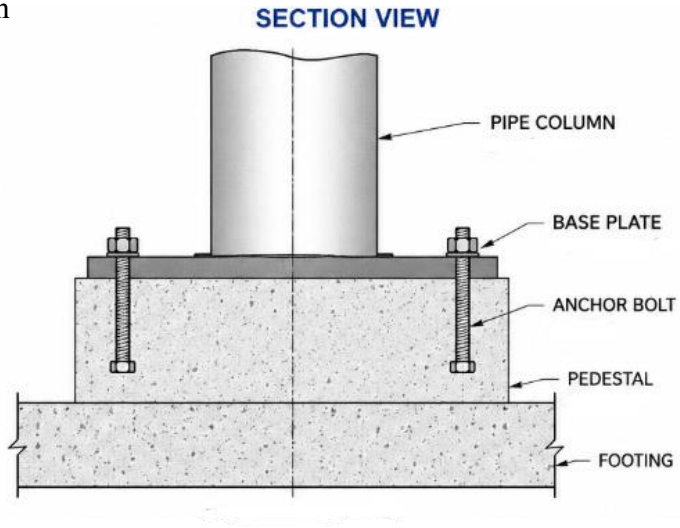
2. Base plate segment

Design Parameter	Symbol	Value	Source
Factored axial load (LRFD)	P_u	7550.979 kips	Main Tank Design (Predefined)
Governing overturning moment (LRFD)	M_u	13656.1295 kip-ft	Main Tank Design (Predefined)
Pipe diameter	D_{pip}	30 in	Main Tank Design (Predefined)
Number of anchor bolts	N_b	4	From Anchor Bolt Design
Anchor bolt diameter	d_b	2 in	From Anchor Bolt Design
Anchor bolt hole diameter	d_b	2.5 in	From Anchor Bolt Design
Bolt arrangement dimensions	b×l	38 in × 12 in	From Anchor Bolt Design
Base-Plate cross-section	D_{co}	40	Main Tank Design (Predefined)
Pedestal cross-section	P×L	54 in × 48 in	From Anchor Bolt Design
Concrete compressive strength	f'_c	4 ksi	User Input
Steel yield strength	F_y	36 ksi	User Input
Modulus of elasticity	E	29,000 ksi	AISC Material Property
LRFD resistance factor (plate bending)	ϕ_b	0.90	AISC LRFD
Required base plate thickness	t_{bp}	To be determined	AISC Base Plate Design



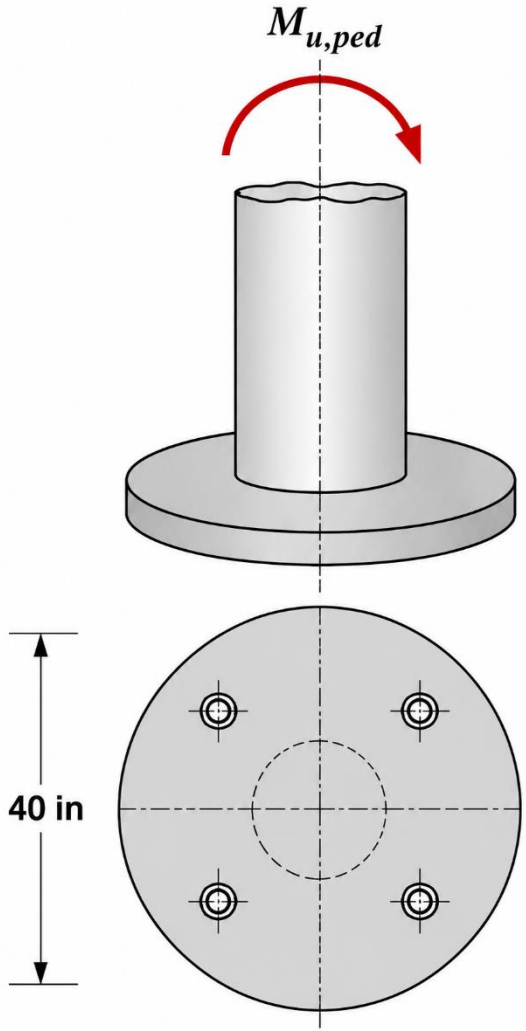


1. Determine the Factored Moment.

$$M_{u,ped} = \frac{M_u}{N_{ped}}$$

$$M_{u,ped} = \frac{13656.1295}{6}$$

$$M_{u,ped} = 2276.022 \text{ kip-ft}$$



Convert to kip-in:

$$M_{u,ped} = 2276.022(12)$$

$$\mathbf{M_{u,ped} = 27312.259 \text{ kip-in}}$$

2. Pedestal Area

$$A_2 = (54)(48)$$

$$A_2 = 2592.0 \text{ in}^2$$

Base Plate Area

$$A_1 = \frac{\pi D^2}{4} = A_1 = \frac{\pi(40)^2}{4}$$

$$A_1 = 1256.637\ in^2$$

Compute Bearing Stress

$$F_p = 0.85\phi_c f'_c \sqrt{\frac{A_2}{A_1}}$$

$$F_p = 0.85(0.65)(4)\sqrt{\frac{2592.0}{1256.637}}$$

$$F_p = 3.174 \text{ ksi}$$

Check Limit

$$1.7\phi_c f'_c = 1.7(0.65)(4) = 4.42 \text{ ksi}$$

$$3.174 < 4.42 \ldots\ldots\ldots\ldots\ldots\ldots \textbf{PASS}$$

3. Assume Base Plate Cross-Section

Adopt:

$$N = B = 40 \text{ in}$$

Base plate area:

$$A_1 = \frac{\pi(40)^2}{4}$$

$$A_1 = 1256.637 \text{ in}^2$$

Check Bearing Capacity

$$P_n = F_p A_1$$

$$P_n = (2.87)(1256.637)$$

$$P_n = 3988.547 \text{ kips}$$

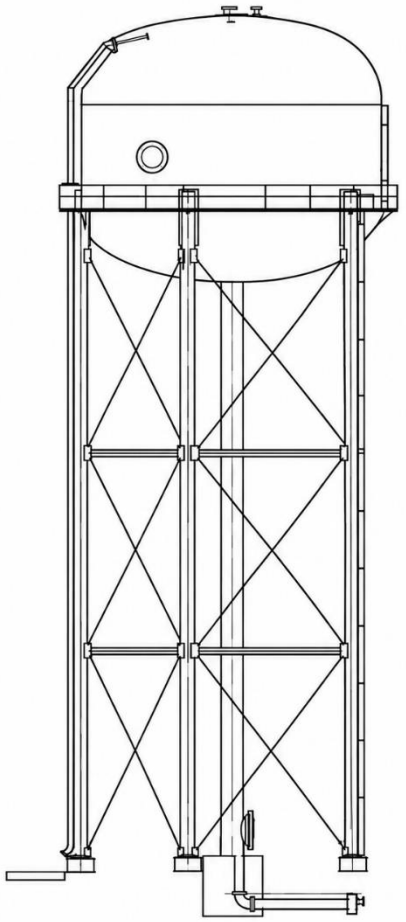
Applied load:

$$Pu,check = 2(Pu,total / Nped)$$

$$P_u = 2516.993 \text{ kips}$$

Note: for this scenario we considered half of the pedestals in compression and half of them in tension as show in the photo

$$3988.547 > 2516.993 \quad \textbf{PASS}$$



4. Determine Equivalent Eccentricity

$$e = \frac{M_{u,Ped}}{P_{leg}}$$

Substitute:

$$e = \frac{27312.259}{1258.5}$$

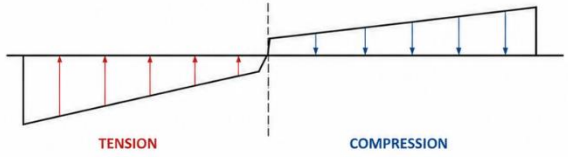
$$e = 21.702 \text{ in}$$

Check Bearing Condition

For full compression:

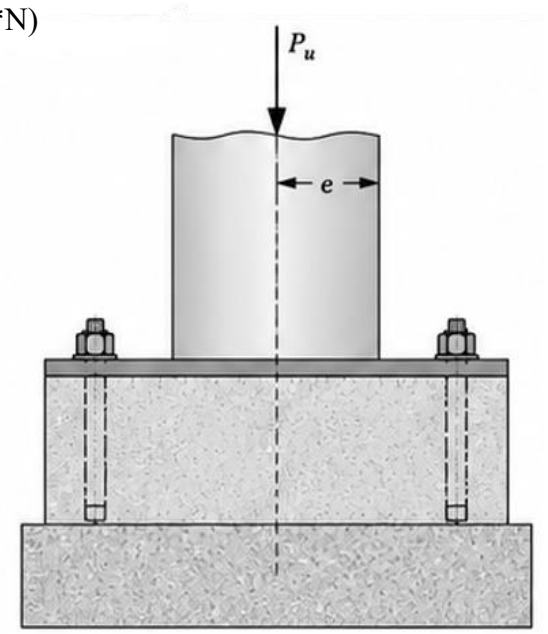
$$e \leq \frac{N}{6}$$

$\frac{N}{6} = \frac{40}{6} = 6.6667$ in……. if this cause controls Pu/(1.0*N)



Since:

$21.702 > 6.6667$……...if this cause controls (Pu/0.5*N)



Full compression does not occur.

Therefore:

- Uplift develops.
- Anchor bolts resist tension.
- Partial compression exists beneath the base plate.

$$\boxed{e = 21.702 \text{ in} > \frac{40}{6} = 6.6667 \text{ in}}$$

5. Determine Resultant Anchor Bolt Force

From the completed anchor bolt design:

$$T_{leg} = 122 \text{ kips}$$

Number of anchor bolts:

$$N_b = 4$$

Anchor force per bolt:

$$T_{bolt} = \frac{T_{leg}}{N_b}$$

$$T_{bolt} = \frac{122}{4} = \mathbf{30.5\ \boldsymbol{kips}}$$

6. Determine Base Plate Thickness

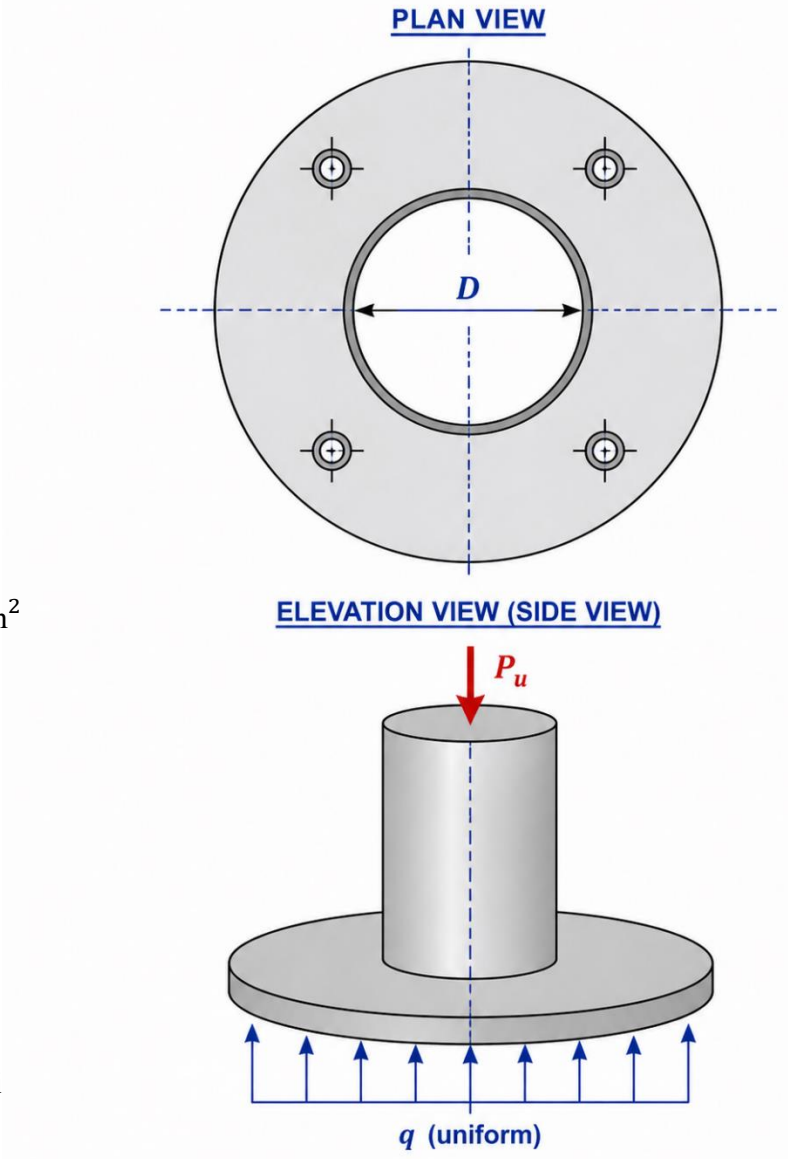
6.1. Determine Bearing Area of Pipe Column

Cylinder diameter:

$$D = 2.5 \text{ ft}$$

$$D = 2.5(12)$$

$$D = 30 \text{ in}$$



Column bearing area:

$$A_b = \frac{\pi D^2}{4}$$

$$A_b = \frac{\pi(30)^2}{4}$$

$$A_b = 706.858 \text{ in}^2$$

6.2. Determine Bearing Pressure

$$q = \frac{P_u}{A_b}$$

$$q = \frac{1258.5}{706.585}$$

$$q = 1.780 \text{ ksi}$$

$$\mathbf{q = 1.780 \text{ ksi}}$$

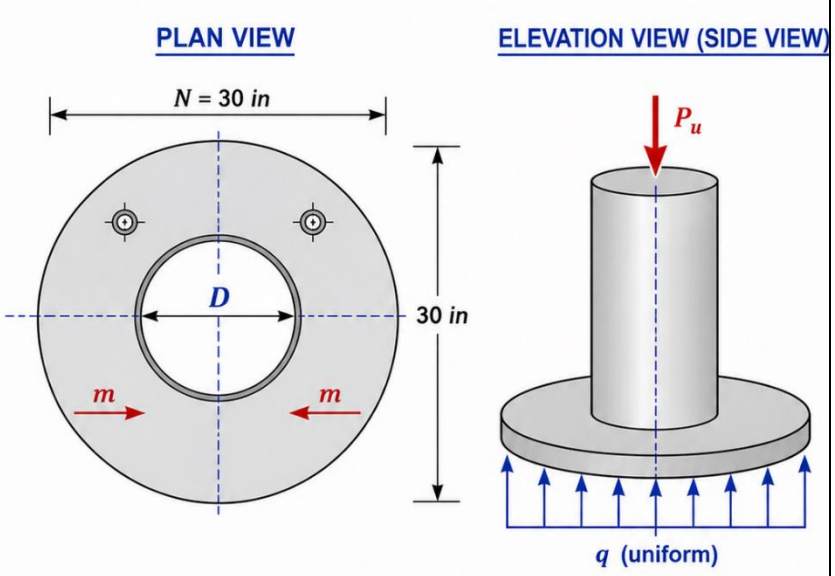
6.3. Determine Plate Projection

$$m = \frac{N - D}{2}$$

$$m = \frac{40 - 30}{2}$$

$$m = 5 \text{ in}$$

$$\mathbf{m = 5 \text{ in}}$$



6.4. Determine Plastic Moment per 1-in Strip

$$M_{plu} = \frac{qm^2}{2}$$

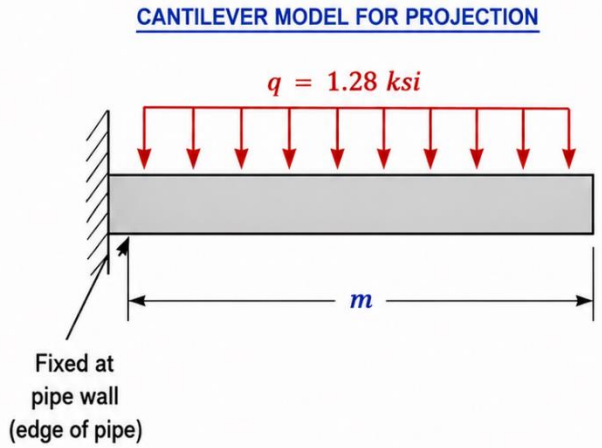
$$M_{plu} = \frac{(1.780)(5)^2}{2}$$

$$M_{plu} = 22.255 \text{ kip-in/in}$$

$$M_{plu} = 22.255 \text{ kip-in/in}$$

6.5. Determine Required Thickness

Using AISC LRFD:



$$t_p = \sqrt{\frac{4M_{plu}}{0.90F_y}}$$

$$t_p = \sqrt{\frac{4(22.255)}{0.90(36)}}$$

$$t_p = \sqrt{\frac{89.02}{32.40}}$$

$$t_p = \sqrt{2.748}$$

$t_p = 1.659$ in $\mathbf{t_p = 2 \text{ in}}$ Adopt: $\underline{\mathbf{t_p = 2 \text{ in}}}$